Sankalp Singh Bais

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https://github.com/bash-sanka1p

Software Engineer

Software Engineer with **1.5+ years** of experience developing real-time edge applications, computer vision solutions, and **cloud infrastructure**. Contributed to a production Raspberry Pi 5 system with object detection, custom OCR, low-latency video streaming, and live location reporting. **Reduced Azure costs 60–70%** through Container Apps migration and helped **maintain 99.9%** uptime via **GitLab CI/CD**. Currently developing a centralized monitoring dashboard using Prometheus, Grafana, and Loki for **23+ remote devices**.

WORK EXPERIENCE

Technical Trainee (Software, Hardware, and DevOps) — 06/2024 - Present

Containova Systems LLP — **Navi Mumbai, India**

- **Optimized** Azure costs by **60-70%** through migration to **Azure Container Apps**, integrating with **Azure Container Registry**.
- **Deployed and managed** scalable Azure and on-premises environments with Dokploy, ensuring **99.9% uptime**.
- **Secured** development workflows by hardening **GitLab** (RBAC + SSL), boosting collaboration efficiency by **30%**
- **Built 3+ GitLab CI/CD pipelines** for automated build, test & rollout
- **Containerized** Express, Flask, and Angular applications to enhance deployment efficiency and scalability.
- **Integrated Azure Container Registry (ACR)** with GitLab CI/CD for auto-publish and zero downtime.
- **Automated** configuration using **10+ Python/Bash scripts**, reducing bare-metal configuration time by **67%**.
- **Developed** Python scripts for a **centimeter-accurate GPS**, enabling precise tracking in logistics applications.
- **Secured** remote and on-prem devices using **VPN** tools, including WireGuard, Zerotier, Tailscale, and Headscale.
- **Designed and deployed** 3+ embedded system prototypes using **Raspberry Pi 5** and **Jetson Nano**.
- **Integrated** ASIC modules for a **300%** boost in computational throughput.
- **Developing** a **monitoring dashboard** using **Grafana** and **Prometheus**, supporting real-time oversight of **23+ remote** devices and **3+ servers**.
- **Built** Flask + Celery + Redis backend with Zoho Cliq notifications; **migrated** from threading to Celery workers for **improved scalability**
- **Repurposed LPRNet** and **trained** on 7,000+ custom images achieving **88.21% accuracy**; replaced **EasyOCR** in edge and backend systems
- **Developed and maintain** real-time **Python application** on Raspberry Pi 5 for object detection, custom OCR, zero-frame-delay RTSP streaming (OpenCV), live GPS socket transmission, and error notifications

Hardware and Software Intern — 07/2023 - 06/2024

Containova Systems LLP — **Navi Mumbai, India**

- **Deployed self-hosted GitLab** on-premises and configured Jetson Nano and **Raspberry Pi 4** environments to meet project specifications.
- **Trained** CNN models and supported high-precision GPS system development.
- **Documented** hardware configuration and deployment process for internal team use.
- **Created HTTP server** to serve high-precision **GPS data**

EDUCATION

B.Tech. Computer Science and Engineering

Shri Shankaracharya Technical Campus • GPA: 8.62 — Bhilai, Durg • 11/2020 - 05/2024

CERTIFICATIONS

DevOps Engineering: Planning To Production 05/2025

Geeks For Geeks

PROJECTS

System Monitoring Dashboard 08/2025 - Present

Developing a system monitoring dashboard using Prometheus, Node Exporter, Loki, and Promtail, with visualization through Grafana.

NORS Automata v1.0 Home Automation 03/2023 - Present

Developed a Raspberry Pi-based home automation system with voice, gesture, and web-app control during a hackathon.

SKILLS

Languages: Bash, Python

Tools: Docker, Dokploy, Git, GitLab, Grafana, Loki, Prometheus

Cloud Platforms: Microsoft Azure (VMs, ACR, ACS, App Service, Blob Storage, Postgres, Log Workspace)

Operating Systems: Linux (Ubuntu, Raspbian)

Embedded Systems: ASIC Integration, Jetson Nano, Proprietary GPS Modules, Raspberry Pi 5

Utilities & Networking: crontab, headscale, nmcli, systemd, tailscale, wireguard, Zerotier

Backend: Celery, Flask, Redis, REST APIs, Socket Programming

Computer Vision & Machine Learning: CNN, EasyOCR, LPRNet, OpenCV